# SSMU-Net: A Style Separation and Mode Unification Network for Multimodal Remote Sensing Image Classification

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***Abstract*— The rapid progress in remote sensing technology has made it convenient for satellites to capture both multispectral (MS) and panchromatic (PAN) images. MS has more spectral information, and PAN has higher spatial resolution. How to exploit the complementarity between MS and PAN images, and effectively combine their respective advantageous features while alleviating mode differences, has become a crucial research task. This article designs a style separation and mode unification network (SSMU-Net) for MS and PAN image classification from a novel and effective perspective. The network can be divided into two stages: style separation and mode unification (MU). In the style separation stage, we use wavelet decomposition and techniques similar to generative adversarial networks to preliminarily separate the information of MS and PAN into different components. These components better preserve complete information from the original data and have their own advantages in style and content. Then, we propose a symmetrical triplet traction module to perform style traction on different components, making style features more unique and content features more unified, achieving feature separation and purification. In the MU stage, we design an encoder–decoder model to reduce the impact of mode differences. The experimental results from multiple datasets validate the effectiveness of our proposed method. Our overall accuracy (OA) improved by approximately 4% on the Shanghai and Beijing datasets, and it has exceeded 99.28% on the Hohhot and Vancouver datasets. Our code is available at: https://github.com/proudpie/SSMU-Net.**

***Index Terms*— Deep learning (DL), fusion classification, multispectral (MS) images, panchromatic (PAN) images, remote sensing.**

## I. Introduction

NOWADAYS, with the rapid development of Earth observation technology, remote sensing images with high spectral information and high spatial resolution can be obtained easily. Many remote sensing satellites, such as IKONOS, QuickBird, Deimos-2, and Gaofen-2 [1], [2], can simultaneously capture panchromatic (PAN) images and multispectral (MS) images, which are widely used in defense security, precision agriculture and biomass estimation [3], [4], [5]. MSs have more spectral information, which is beneficial for identifying land cover. However, due to sensors' technological constraints, the spatial resolution of MSs is usually lower than that of PANs. Therefore, exploiting the intrinsic complementarity between the spectral information in MS images and the spatial information in PAN images has gained an increasing interest in remote sensing image classification [6], [7]. In order to effectively combine the information of MS and PAN, many fusion methods have been proposed, which can be roughly divided into two types: the methods involving image data fusion first and then classification [8], [9], [10], [11], and the methods involving the fusion of features extracted from MS and PAN images followed by classification [12], [13].

The former type mainly uses high-resolution PAN images to sharpen low-resolution MS images to generate fused high-resolution MS images, and then extract features for classification. Recent years have witnessed various algorithms coming to the fore based on the following well-known methods: component substitution (CS) (including intensity-hue-saturation (IHS) transform [14], [15], Gram–Schmidt (GS) [16], [17], Brovey transform (BT) [18], and principal component analysis (PCA) [19], [20], [21]); multiresolution analysis (MRA) (including discrete wavelet transform (DWT) [22], [23], [24], and Laplace pyramid transform (LPT) [25], [26]); variational optimization (VO) [27]; and sparse representation (SR) [28], [29], [30]. Shackelford and Davis [10] extracted different features from data sources, using fuzzy logic methods for classification. Jiang and Wang [31] separated images into cartoon and texture components based on morphological component analysis and SR and essentially fused the representation coefficients to obtain higher quality images. Liu et al. [32] produced high-quality PAN-sharpened images from the perspective of generative adversarial learning. While the aforementioned algorithms can quickly and effectively obtain fusion results, they often rely on pansharpening, which is often challenging to avoid detail loss and high-frequency spectral distortion.

The latter type first extracts features from MS and PAN, respectively, and then fuses them for classification. With the rapid development of deep learning (DL), this method

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has gradually become a popular research field in remote sensing [33], [34], [35], [36], [37], [38], [39], [40], [41]. Zhu et al. [33] introduced an adaptive center–offset sampling strategy (ACO–SS) and designed separate attention modules for MS and PAN. Tian et al. [37] constructed an interpretable variational sharpening deep network by expanding the variable splitting method, achieving an excellent balance between spatial and spectral quality. Ma et al. [38] adopted an adaptive dilation rate selection strategy (ADR-SS) instead of traditional convolution to solve the problem of large-scale differences in category regions, and migrated the central pixel to the location with the most similar neighborhood information for classification. Different modules were designed for each feature extraction stage of the two branches, and collaborative progressive fusion modules were designed to eliminate differences. Zhou et al. [41] developed a customized transformer architecture that formulates the features of PAN and MS into queries and keywords to facilitate multimodal joint learning and then used invertible neural modules for feature fusion.

The aforementioned explorations have led to significant contributions to remote sensing image fusion and classification, providing valuable insights for designing modules. However, there remain certain issues that have been disregarded and require further attention.

1) MS and PAN images share consistent content information for the same area and contain unique style information in both spatial and spectral domains. Many studies usually directly learn strong mapping relationships between multiple modes through a large amount of raw remote sensing data, or simply consider designing attention modules and collaborative strategies for classification. The correlation and differences between MS and PAN content information and style information have rarely been considered. This oversight leads to confusion and redundancy between content information and style information, hindering the accurate modeling of intrinsic features and relationships. Consequently, many key semantic features become obscured, reducing the fusion effect and affecting classification performance.

2) Due to the diverse acquisition methods and technical parameters employed for MS and PAN images, they exhibit a distinctive emphasis on their respective spectral and spatial domains. This leads to significant mode differences between the two images. Mode differences can be described as having different feature representations for the same scene information, which makes it difficult for features to match and effectively fuse during the fusion process. However, current algorithms rarely pay attention to this issue, leaving room for optimization.

Based on the above problems, we propose a style separation and mode unification network (SSMU-Net) for MS and PAN image classification. The contributions of our work can be summarized as follows.

1) We propose a style separation (SS) strategy that utilizes wavelet decomposition and generative adversarial networks to obtain different style feature components that are not limited to low-level features. This provides favorable prerequisites for the traction of features and protects advantageous features conducive to classification. We introduce two sets of triplet losses to guide the development of different styles and content features. This optimization process allows us to obtain more comprehensive, richer, and purer feature representations. The whole strategy helps us better capture the intrinsic features and relationships of data, improves the utilization of features, and contributes to the effectiveness of fusion.

2) To reduce the interference caused by mode differences in feature fusion, we propose a mode unification (MU) module. We train two independent encoder–decoder models to process features from different modes and map the features to an intermediate mode with a similar feature representation. The intermediate mode can ensure all features exhibit similar feature representations, solving the fusion problem caused by mode differences while retaining advantageous information about each mode. Through this module, different features in different modes can be better fused.

The remainder of this article is structured as follows. Section II provides an overview of related work. Section III presents a detailed description of the proposed method. Section IV presents the experimental setup, including the datasets used and the evaluation criteria, and shows the experimental results and corresponding analysis. Finally, Section V summarizes the findings and concludes the article.

## II. Related Work

In this section, we provide an introduction to three works related to our method, including wavelet decomposition, triplet loss, and generative adversarial network.

### A. Wavelet Decomposition

Wavelet decomposition [22] is a widely used method for preprocessing remote sensing images. It utilizes high-pass and low-pass filters to decompose an image into a set of low-frequency (representing smooth variations) and high-frequency subbands (representing edges, textures, and details). This decomposition separates different frequency information of the image effectively for representation and analysis.

One common approach to wavelet decomposition in remote sensing is the DWT [42], whose formula can be expressed as follows:

$$W_{j,k}(x, y) = \sum_{m1=0}^{N-1} \sum_{m2=0}^{N-1} h_{m1} \cdot h_{m2} \cdot X_{j-1,m1,x} \cdot X_{j-1,m2,y} \quad (1)$$

where $W_{j,k}(x, y)$ represents the coefficient of the $j$th level DWT at position $(x, y)$, $h_{m1}$ and $h_{m2}$ are the discrete wavelet coefficients of the row and column filters, respectively. $X$ denotes approximate coefficients of the original image in level $j - 1$.

Overall, wavelet decomposition, as a preprocessing step for remote sensing images, is very useful in improving image quality and facilitating analysis [43].

### B. Triplet Loss

Triplet loss [44] is a loss function for training DL models for similarity measurement or feature embedding. The goal

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of triplet loss is to minimize the distance between the anchor sample and the positive sample while maximizing the distance between the anchor sample and the negative sample. The formula for triplet loss is as follows:

$$L = \max(0, m + d(a, p) - d(a, n)) \tag{2}$$

where $m$ is a constant controlling the distance threshold between positive and negative samples, $d(a, p)$ is the distance between the anchor sample and the positive sample, and $d(a, n)$ is the distance between the anchor sample and the negative sample. If the distance between the anchor sample and the positive sample is small, the loss is small, and vice versa.

### C. Generative Adversarial Network (GAN)

The core idea of GAN [45] is to use the adversarial loss function to drive the generator to generate data similar to real data.

Generator $G$ takes random noise $z$ as input and generates a fake image $G(z)$, while discriminator $D$ takes it as input and outputs a real number belonging to [0,1], indicating the probability that the image is a real image. $D$ tries to maximize the probability of correctly classifying real images as 1 and false images as 0, while $G$ tries to minimize $1 - D(G(z))$ to make $D$ unable to distinguish whether $G(z)$ is a real image.

The loss function of GAN is called adversarial loss, which can be expressed as

$$\min_G \max_D V(D, G) = \mathbb{E}_{x\sim p_{\text{data}}(x)}\left[\log D(x)\right] + \mathbb{E}_{z\sim p_z(z)}\left[\log(1 - D(G(z)))\right] \tag{3}$$

where $p_{\text{data}}(x)$ represents the true data distribution and $p_z(z)$ represents the noise distribution. The training of GAN is a process of mutually optimizing $G$ and $D$.

## III. Proposed Method

### A. SS strategy

To deal with the first issue in Section I, we propose the SS strategy to separate and guide the extraction of both content and style information from $MS \in \mathbb{R}^{W\times H\times 4}$ and $PAN \in \mathbb{R}^{4W\times 4H\times 1}$. SS strategy has three parts, including wavelet decomposition and image reconstruction, producing style components by generative adversarial networks, and the symmetric triplet traction (STT) module. The overall process of the SS strategy is shown in Fig. 1.

*1) Wavelet Decomposition and Image Reconstruction:* The SS strategy we design requires the traction of features obtained from different branches. Therefore, we first need to obtain inputs of content and style branches from *MS* and *PAN*.

Wavelet decomposition [22] is very suitable for our needs. In 2-D wavelet decomposition, the image is decomposed into multiple subbands of different scales and orientations. The low-frequency subband contains the smoother information in the image, while the high-frequency subband contains high-frequency details and textures. This decomposition is very close to the separation of content and style information of images. Since we only need to obtain rough initial inputs in this part, we use the simple and fast Harr wavelet [46]. The process of wavelet decomposition and image reconstruction is shown in Fig. 2.

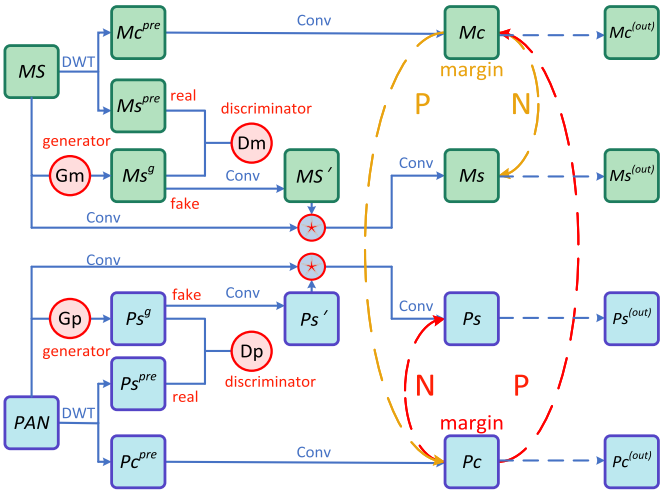



Fig. 1. Overall structure of the SS strategy. Images labeled "*pre*" on names are reconstructed images obtained from the first part. Images labeled "*g*" are images obtained by generators in the second part. Curves of the same color are connected to a set of triples, and through the traction of two sets of triples, we obtain outputs of the third part, which are also outputs of the entire strategy. "Conv" represents 2-D convolution. The symbol "⊛" denotes element-wise matrix multiplication, where each element in one matrix is multiplied with the corresponding element in another matrix.

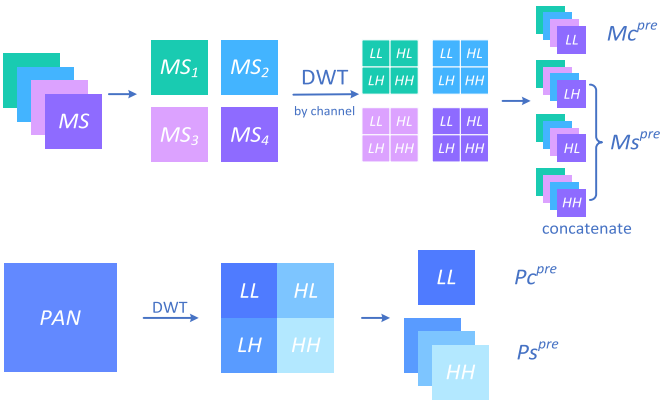



Fig. 2. Schematic of image wavelet decomposition and reconstruction. Wavelet decompositions are performed on each channel of *MS*. Each time the wavelet reduces the height and width of the image by half. The low-frequency results (*LL*) are used as inputs for the content branch, while the remaining high-frequency results (*LH, HL*, and *HH*) are used as inputs for the style branch. The size adjustment is handed over to the subsequent deep network.

We perform a single-level wavelet decomposition on each channel of MS

$$M_i_LL, M_i_LH, M_i_HL, M_i_HH = \text{DWT}(MS_i) \tag{4}$$

where $i$ represents the $i$th channel of *MS* and DWT represents discrete wavelet transform.

Then, we concatenate $M_i_LL$ to $Mc^{pre}$ by channel and concatenate $M_i_LH$, $M_i_HL$, $M_i_HH$ to $Ms^{pre}$ by channel. In this case, $Mc^{pre} \in \mathbb{R}^{0.5W\times 0.5H\times 4}$ and $Ms^{pre} \in \mathbb{R}^{0.5W\times 0.5H\times 12}$. $Mc^{pre}$ and $Ms^{pre}$ represent rough initial content and style

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components of *MS*. The specific equations are as follows:

$$\begin{aligned} M_LL &= M_1_LL \oplus M_2_LL \oplus M_3_LL \oplus M_4_LL \\ M_LH &= M_1_LH \oplus M_2_LH \oplus M_3_LH \oplus M_4_LH \\ M_HL &= M_1_HL \oplus M_2_HL \oplus M_3_HL \oplus M_4_HL \\ M_HH &= M_1_HH \oplus M_2_HH \oplus M_3_HH \oplus M_4_HH \end{aligned} \tag{5}$$

$$\begin{aligned} Mc^{pre} &= M_LL \\ Ms^{pre} &= M_LH \oplus M_HL \oplus M_HH \end{aligned} \tag{6}$$

where $\oplus$ represents channel concatenation.

Similarly, we perform wavelet decomposition on *PAN*

$$P_LL,\ P_LH,\ P_HL,\ P_HH = \mathrm{DWT}(PAN). \tag{7}$$

The low-frequency component $P_LL$ is taken as the initial content component of *PAN* ($Pc^{pre}$), while $Ps^{pre}$ is made by concatenating the other three components. $Pc^{pre}$ and $Ps^{pre}$ represent rough initial content and style components of *PAN*. In this case, $Pc^{pre} \in \mathbb{R}^{2W\times 2H\times 1}$ and $Ps^{pre} \in \mathbb{R}^{2W\times 2H\times 3}$

$$\begin{aligned} Pc^{pre} &= P_LL \\ Ps^{pre} &= P_LH \oplus P_HL \oplus P_HH. \end{aligned} \tag{8}$$

The results of wavelet decomposition have clear directional attributes, providing the point of entry for the subsequent implementation of achieving style separation and traction between *MS* and *PAN*. Through the above operations, we obtain the rough content and style components of *MS* and *PAN* ($Ms^{pre}$, $Mc^{pre}$, $Ps^{pre}$, and $Pc^{pre}$). Providing preliminary trends for each branch, they are only used as the basic inputs and are further processed in subsequent operations. Thus, we do not need to have particularly precise requirements for their quality.

*2) Producing Style Component by GAN:* In this module, we continue to optimize the initial inputs. The specific operation is as follows.

We adopt a strategy similar to GAN [45], using convolution to generate higher quality style features by adversarial loss. The essence of GAN is to use the adversarial loss to make generated results similar to real results, so we only use it to optimize the effectiveness of our networks' feature extraction. As there are sources that can extract features, we use the original images (*MS* and *PAN*) instead of the original input of GAN (random noise).

First, we extract features of *MS* and *PAN* and simulate the generator of GAN to generate images of the same size as $Ms^{pre}$ and $Ps^{pre}$, respectively. The generated outputs closely resemble $Ms^{pre}$ and $Ps^{pre}$ but are not exactly the same for the reason of taking both shallow and deep features into account

$$\begin{aligned} Ms^{gan} &= \mathrm{Gm}\left(MS; w_m^g\right) \\ Ps^{gan} &= \mathrm{Gp}\left(PAN; w_p^g\right) \end{aligned} \tag{9}$$

where Gm and Gp are generators and $w_m^g$ and $w_p^g$ are parameters that need to be learned. Superscript $g$ represents the generator, subscript $m$ and $p$ represent *MS* and *PAN*. Throughout the following of this article, symbols $w_*^*$ denote learnable parameters.

Second, taking the MS style branch as an example, considering $Ms^{pre}$ as real samples and $Ms^{gan}$ as fake samples, we construct a discriminator Dm. The output of Dm is a probability value that belongs to [0,1]. We design the loss function as a binary cross-entropy loss to optimize the training of Dm and Gm. Similarly, we perform the same treatment on $Ps^{pre}$ and $Ps^{gan}$. The specific loss function of each network is as follows:

$$\begin{aligned} &L^{\mathrm{Dm}} \\ &= -\mathbb{E}\log\left(\mathrm{Dm}\left(Ms^{pre}; w_m^d\right)\right) - \mathbb{E}\log\left(1 - \mathrm{Dm}\left(Ms^{gan}; w_m^d\right)\right) \\ &L^{\mathrm{Dp}} \\ &= -\mathbb{E}\log\left(\mathrm{Dp}\left(Ps^{pre}; w_p^d\right)\right) - \mathbb{E}\log\left(1 - \mathrm{Dp}\left(Ps^{gan}; w_p^d\right)\right) \end{aligned} \tag{10}$$

$$\begin{aligned} &L^{\mathrm{Gm}} \\ &= -\mathbb{E}\log\left(\mathrm{Dm}\left(Ms^{gan}; w_m^d\right)\right) \\ &L^{\mathrm{Gp}} \\ &= -\mathbb{E}\log\left(\mathrm{Dp}\left(Ps^{gan}; w_p^d\right)\right) \end{aligned} \tag{11}$$

where superscript $d$ represents the discriminator.

Throughout the learning process, discriminators gradually use the style of input images as the criterion for discrimination, while generators' extraction focuses on approaching the images' style constantly. The style features obtained through the first part serve as the learning tendency of the network. However, we do not require the generated results to be completely consistent with real samples. The generated images are allowed to retain information from the original image source.

Through this section, we achieve more reasonable style features by using the adversarial loss. This section has two advantages. First, compared to wavelet decomposition, the generated results contain more information from the original data source. After wavelet decomposition, the information content of high-frequency components is severely insufficient, making it hard to undertake subsequent style traction operations. Especially, the decomposition results of *PAN* almost only retain blurry contours. Our operation alleviates the problem of insufficient high-frequency information. Second, these features can better balance the shallow and deep information, and retain more semantic information. We also verify the rationality of this method in Section IV.

*3) STT Module:* Since *MS* and *PAN* depict the same scene, their content branches should be similar, and their unique style branches should be separated. Therefore, we propose the STT module to use the triplet loss [44] to further widen the differences between style features ($Ms^{gan}$ and $Ps^{gan}$) of $MS$ and $PAN$ and to make the content features ($Mc^{pre}$ and $Pc^{pre}$) of $MS$ and $PAN$ close to each other. The specific structure of STT is shown in Fig. 3.

*Process of Style Components:* In this module, we need to use the triplet loss as the traction for output features. The premise of using triplet loss is that the information contained in each branch should be in the same order of magnitude. However, the information on style branches obtained from previous operations is still insufficient. If they directly participate in the

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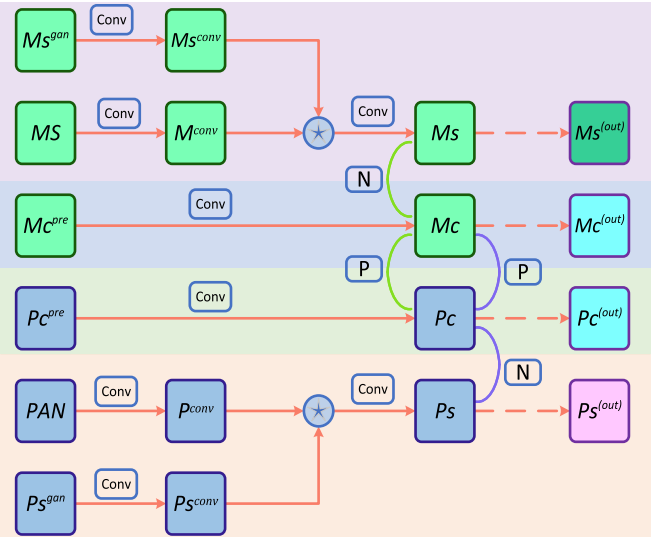



Fig. 3. Struture of STT module. *Ms*, *Mc*, *Ps*, and *Pc* are in the same size. All convolutional layers update parameters simultaneously under the influence of two pairs of triplet loss. Curves with the same color represent the same set of triplet data.

measurement of distance and the traction of the branches, serious feature destruction will happen easily. Therefore, we use style features obtained in the previous operation ($Ms^{gan}$ and $Ps^{gan}$) as masks to enhance the advantageous information of features extracted from original images

$$\begin{aligned} Ms &= \mathrm{Conv}(MS; w_m) \cdot \sigma(\mathrm{Conv}(Ms^{gan}; w_{ms})) \\ Ps &= \mathrm{Conv}\big(PAN; w_p\big) \cdot \sigma\big(\mathrm{Conv}\big(Ps^{gan}; w_{ps}\big)\big) \end{aligned} \tag{12}$$

where $\sigma$ represents the sigmoid function.

$Mc^{pre}$ and $MS$ have a similar information level. As $Ms$ is obtained by convolving the original image and multiplying it by a mask, its information level is similar to the information level of the content component. The larger the value of masks, the more style information needs to be considered, and vice versa. By the method of using masks, $Ms$ is more focused on style features. Similarly, we process $Ps^{gan}$ and $PAN$ in the same way to obtain $Ps$.

*Process of Content Components:* Simultaneously, we convolve $Mc^{pre}$ and $Pc^{pre}$ to obtain $Mc$ and $Pc$. To facilitate the distance measurement of triplet loss, we integrate all components into the same size by convolution

$$\begin{aligned} Mc &= \mathrm{Conv}(Mc^{pre}; w_{mc}) \\ Pc &= \mathrm{Conv}\big(Pc^{pre}; w_{pc}\big). \end{aligned} \tag{13}$$

*Design of Triplet Loss:* In the design of the triplet loss, we want the content and style features of each source to be as far away as possible, while the content features of $MS$ and $PAN$ should be as close as possible. Therefore, we design two sets of symmetric triplet loss. In the first set of triplet loss, we appoint $Mc$ as the anchor and $Pc$ as the positive sample to get close to $Mc$ and appoint $Ms$ as the negative sample to stay away from $Mc$. Similarly, in the second set of triplet loss, we appoint $Pc$ as the anchor and $Mc$ as the positive sample to approach $Pc$, and appoint $Ps$ as the negative sample to stay away from $Pc$. We call these two sets of triplet loss $L_1(Mc, Pc, Ms)$ and $L_2(Pc, Mc, Ps)$. The calculation formulas of the loss function are as follows:

$$\begin{aligned} L_1(Mc, Pc, Ms) = \max\big\{&\mathrm{Avg}\big(||Mc_i - Pc_i||_2^2\big) \\ &- Avg\big(||Mc_i - Ms_i||_2^2\big) + \alpha, 0\big\} \end{aligned} \tag{14}$$

$$\begin{aligned} L_2(Pc, Mc, Ps) = \max\big\{&\mathrm{Avg}\big(||Pc_i - Mc_i||_2^2\big) \\ &- Avg\big(||Pc_i - Ps_i||_2^2\big) + \alpha, 0\big\} \end{aligned} \tag{15}$$

where $\alpha$ is a hyper-parameter used as margins and $i$ represents the channel number.

In the original usage of triple loss, the goal is to learn a deep embedding that maps faces to high-dimensional space, where similar faces are closer and different faces are farther apart. In our approach, we use the triplet loss function to learn a filter that maps input features from different branches to features that better match ideal styles. The primary objective is to obtain high-quality features to facilitate final fusion and classification. Therefore, the selection of the anchor, positive, and negative samples is not arbitrary. Remote sensing data only has content consistency, not style consistency. The advantage of using content features as anchors is that another branch of content features (positive samples) can be found to approach each other, while actively maintaining differences from style features (negative samples) from the same data source. We do not use style features as anchors for the reason that it is difficult to select suitable positive samples and obtain the traction manner required to complete our task.

*Cross-Entropy Loss for Traction:* As mentioned above, the goal of the STT module is not to train mappings but to obtain high-quality features. However, the strengthening style may not fully align with our goal, as the direction of style traction may not be consistent with the classification task. Therefore, we connected a classification network after this module. We consider the cross-entropy loss ($L_c$) of classification as a part of the module's loss, which helps guide features in a direction conducive to classification. The final loss can be expressed as

$$L^{\text{total}} = \lambda_1 \times (L_1 + L_2) + \lambda_2 \times L_c \tag{16}$$

where $\lambda_1$ and $\lambda_2$ are factors to balance the multiple objectives.

Generally, under the impact of the entire SS strategy, we separate style and content branches from the original $MS$ and $PAN$ and make the focus of each branch more prominent through the traction of triplet loss, achieving preliminary feature purification.

### B. MU Module

Given the inherent characteristics of remote sensing images, although mode differences between $MS$ and $PAN$ provide rich information for the extracted style features, they increase the differences in feature representation. This difference worsens the effectiveness of direct feature fusion [6]. In order to achieve better feature fusion, we need to transform the features of each branch from different modes to an intermediate mode that is conducive to fusion. Therefore, we propose a MU module to project features from different modes onto a suitable intermediate mode, enabling features to be fused effectively. The whole process of the MU module is shown in Fig. 4.

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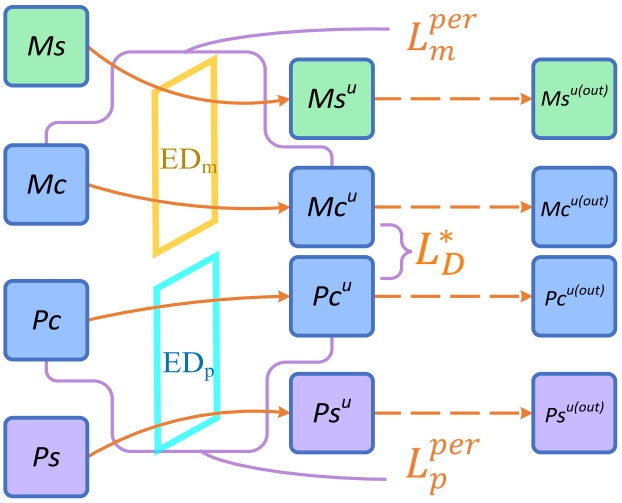



Fig. 4. Structure of MU module. During the training process, $Mc$ and $Pc$ are first processed. Perceptual loss and MSEloss ($L_D^*$) are calculated by output and then used to optimize encoders. After the training is completed, $Mc$, $Ms$, $Pc$, and $Ps$ are simultaneously encoded and unified into the intermediate mode.

*1) Input and Output of Content Branches:* There are still differences between $Mc$ and $Pc$. Since the content information they contain is ideally consistent, the differences here are more likely to come from mode differences, which can be understood as the difference in feature representation from different sources. We train encoder–decoder models (abbreviated as ED) to eliminate this mode difference

$$\begin{aligned} Mc^u &= \mathrm{EDm}(Mc; w_{cm}) \\ Pc^u &= \mathrm{EDp}\big(Pc; w_{cp}\big) \end{aligned} \tag{17}$$

where EDm is used to recode features extracted from $MS$ to the intermediate mode, and EDp is used to recode features extracted from $PAN$.

*2) Network Optimization:* Then, we measure the distance between $Mc^u$ and $Pc^u$ and try to eliminate it as much as possible to make $Mc^u$ and $Pc^u$ completely consistent through network optimization, as they theoretically have the same content and the way of feature representation

$$L_D = ||Mc^u - Pc^u||_2^2. \tag{18}$$

*3) Improving Loss Function:* During the training process, we need to avoid situations where the encoder focuses so much on reducing distance $Lc$ that it eliminates all features that contribute to classification. Therefore, we also need to use other loss functions to protect the retention of effective information as much as possible. The specific loss function design includes two parts of loss, one is the distance loss ($L_D^*$) between $Mc^u$ and $Pc^u$, and the other is the perceived change of features after network processing ($L_m^{per}$ and $L_p^{per}$). Perceived loss preserves mid and high-level information about features, ensuring that features required for final classification are not lost

$$L_D^* = \max\big\{||Mc^u - Pc^u||_2^2 - \theta, 0\big\} \tag{19}$$

$$L_m^{\mathrm{per}} = ||F_l\big(Mc^u\big) - F_l(Mc)||_2^2 \tag{20}$$

$$L_p^{\mathrm{per}} = ||F_l\big(Pc^u\big) - F_l(Pc)||_2^2 \tag{21}$$

where $\theta$ is a hyper-parameter to maintain unique useful information for classification in the process of coding, and $F_l(\cdot)$ denotes the output of the $l$th layer of the VGG-16 model.

Finally, the loss calculation formula for entire encoders is as follows:

$$L_{MU} = \lambda^{\mathrm{dis}} \times L_D^* + \lambda^{per} \times \big(L_m^{per} + L_p^{\mathrm{per}}\big) \tag{22}$$

where $\lambda^{dis}$ and $\lambda^{\mathrm{per}}$ are factors to balance the loss.

*4) Processing Style Branches:* After training, we project content features into one mode. Then we get EDm used to process features from $MS$ and encoder EDp used to process features from $PAN$. Finally, we use the trained model to process style branches

$$\begin{aligned} Ms^u &= \mathrm{EDm}(Ms; w_{EDm}) \\ Ps^u &= \mathrm{EDp}\big(Ps; w_{EDp}\big). \end{aligned} \tag{23}$$

There are still differences between $Ms^u$ and $Ps^u$. The differences here are inherent style differences in different modes. Besides, $Mc^u$ and $Pc^u$ are quite similar.

Overall, in this module, we project features from different modes onto a suitable intermediate mode that is conducive to fusion.

### C. Construction of SSMU-Net

In this section, we briefly sort out the entire process of our proposed SSMU-Net. The architecture of the SSMU-Net is shown in Fig. 5 and can be divided into the following four sections.

*1) Data Acquisition:* The premise for style traction is to obtain the rough style and content components of $MS$ and $PAN$. We use wavelet decomposition and GAN to achieve it. Through this operation, we obtain the inputs of four branches.

*2) Feature Traction:* We perform feature traction on four branches by triplet loss. Content features become more similar, while style features become more distinctive.

*3) Mode Unification:* After feature traction, we uniformly address mode differences of features. By designing encoders to operate style and content branches separately, we unify the modes of the four branches into an intermediate mode to reduce the pressure on the final fusion classification network.

*4) Fusion Classification:* Due to little difference in the obtained content features, direct fusion may cause slight information redundancy. Thus, we compare $Mc^u$ and $Pc^u$ pixel by pixel and take a larger value to integrate them into a content feature $MPc^u$, which has the same size as $Mc^u$ and $Pc^u$

$$MPc^u = \max\big\{Mc^u, Pc^u\big\}. \tag{24}$$

Finally, we fuse $MPc^u$, $Ms^u$, and $Ps^u$ and put the final feature into a classification network. After several layers of convolution, a global pooling layer, and several fully connected layers, the class probability of each pixel is ultimately estimated

$$L_{final} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C} y_{ij}\log\big(p_{ij}\big) \tag{25}$$

where $N$ is the number of samples, $C$ is the number of categories, $y_{ij}$ is a $0-1$ variable used to indicate whether

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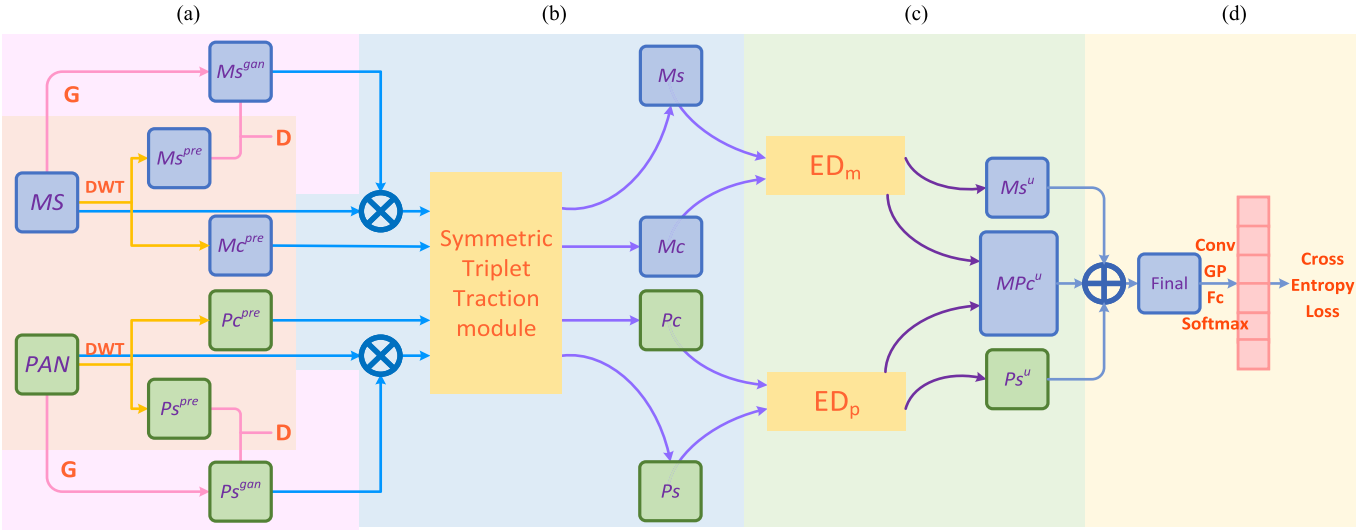



Fig. 5. Overall structure of our network, including (a) data fusion, (b) feature traction, (c) MU, and (d) fusion classification.

sample $i$ belongs to category $j$, and $p_{ij}$ is the probability that the model predicts sample $i$ belongs to category $j$.

## IV. Experimental Results

### A. Datasets Description and Experimental Environment

In this section, we use four datasets to validate the reasonableness and effectiveness of our proposed methods. The details of the datasets are shown as follows.

*1) Hohhot:* Fig. 6(a) shows the Hohhot dataset, with the MS image consisting of 2001 × 2101 × 4 pixels with a spatial resolution of 3.2 m and the PAN image composed of 8004 × 8404 × 1 pixels with a spatial resolution of 0.8 m. The dataset is divided into 11 categories, including water, road, bare land, vegetation, farmland, and six kinds of building areas.

*2) Vancouver:* Fig. 6(b) shows the Vancouver dataset, with the MS image consisting of 873 × 1311 × 4 pixels with a spatial resolution of 4 m and the PAN image composed of 3492 × 5244 × 1 pixels with a spatial resolution of 1 m. The dataset is provided by the 2016 IEEE GRSS Data Fusion Contest, captured by the DEIMOS-2 satellite. It is comprised of eight different classes, namely water, road, tree, vegetation, boat, and three kinds of building areas.

*3) Beijing:* Fig. 6(c) shows the Beijing dataset, with the MS image consisting of 6905 × 7300 × 4 pixels with a spatial resolution of 3.2 m and the PAN image composed of 27 620 × 29 200 × 1 pixels with a spatial resolution of 0.8 m. The dataset can be categorized into ten classes, including water, road, bare land, two kinds of vegetation with different heights, and five kinds of buildings.

*4) Shanghai:* Fig. 6(d) shows the Shanghai dataset, with the MS image consisting of 6905 × 7300 × 4 pixels with a spatial resolution of 3.2 m and the PAN image composed of 27 620 × 29 200 × 1 pixels with a spatial resolution of 0.8 m. The dataset has ten categories, including water, road, bare land, two species of vegetation with different heights, and five kinds of buildings.

The experiments in this article are running on a Workstation with RTX3090Ti 24 GB GPU and 128 GB RAM under Ubuntu 16.04 LTS. The proposed network is trained on PyTorch.

### B. Experimental Setup

In order to evaluate classification performance, we used overall accuracy (OA), average accuracy (AA), and kappa statistic (Kappa) for quantitative analysis of our work. Table I shows the detailed network parameters of the proposed SSMU-Net. In Table I, $c$ represents the number of channels, usually taken as 32, 64, 128, and 256. For the convenience of the display, we omit the cells for stride and padding. We use different strides to unify the size of features, with padding typically being 1 (3 × 3 kernel size) and the rest being 0.

Due to the 4:1 resolution ratio between PAN and MS, we set the input size of the MS image and PAN image patch to be 16 × 16 × 4 and 64 × 64 × 1, respectively. In the training of the networks, we randomly select 2% of the labeled data from each category as training samples. Especially, to alleviate the impact of sample imbalance, we set a sampling threshold of 1500. That is to say, when the size of 2% of samples in a certain category is less than 1500, we select 1500 samples for training. In addition, when selecting test samples, we use the intersection union (IOU) ratio as the sampling standard for testset, which means samples with an IOU ratio greater than 1/3 with other training samples are not included.

We compare our method with five state-of-the-art methods, which are spatial-channel progressive fusion residual network (SCPF-ResNet) [35], spatial-channel collaborative attention network (SCCA-Net) [39], adaptive hybrid fusion network (AHF-Net) [36], multiscale progressive collaborative attention network (MPCA-Net) [38], and transfer representation learning fusion network (TRLF-Net) [34], respectively. To ensure the fairness of the results, all hyper-parameters in the comparison experiments are the optimal parameters in their original paper. In order to reduce the randomness of experimental results and increase the reliability, we repeat each experiment five times and take average values as final results.

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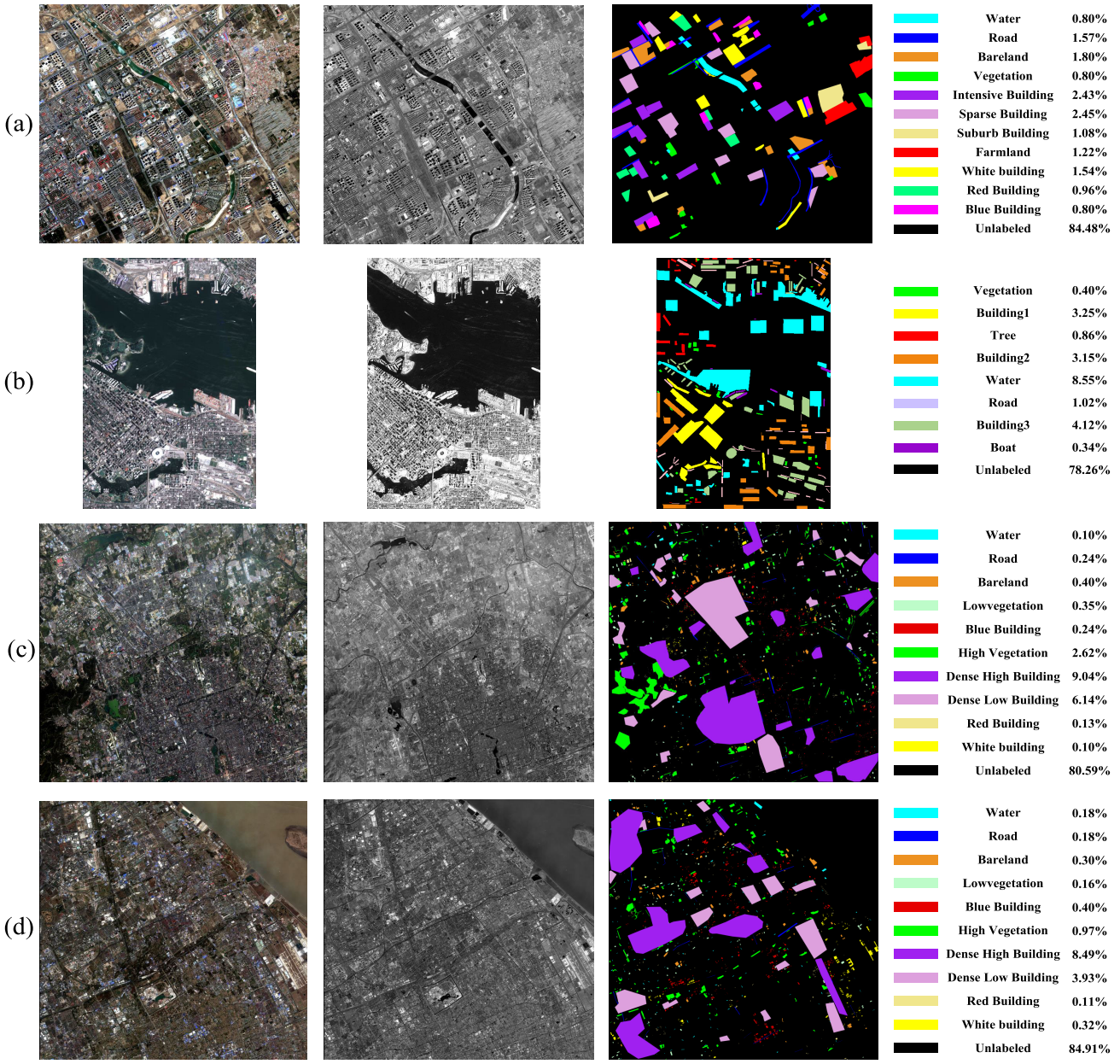



Fig. 6. Each row contains a dataset: (a) Hohhot dataset, (b) Vancouver dataset, (c) Beijing dataset, and (d) Shanghai dataset. (First column) images, (second column) PAN images, and (third column) class labels corresponding to ground truth images. PAN images are adjusted to the same size as MS images for the convenience of display. We use c1,..., c$n$ to represent each category ($n$ represents the number of categories).

### C. Comparison and Analysis of Hyper-Parameters

In this section, we mainly make a detailed comparison and analysis of two important hyper-parameters in our network: the margin $\alpha$ in the SST module and $\theta$ in (19) in the MU module. Because the training of each module is independent, we use the control variable method to analyze the selection of hyper-parameters.

First, with $\theta$ as the middle value 0.3, we vary $\alpha$ from 0 to 0.5 with a stride of 0.1, and obtain the corresponding results in Fig. 7(a). It can be seen that in each dataset, the performance is best when $\alpha$ equals 0.3, and when $\alpha$ is away from 0.3, OA significantly decreases. When $\alpha$ is too small, the distance between the anchor point and the positive and negative samples is too similar, making it difficult to achieve good style separation. When $\alpha$ is too large, the features may be excessively pulled, thereby affecting the classification effect.

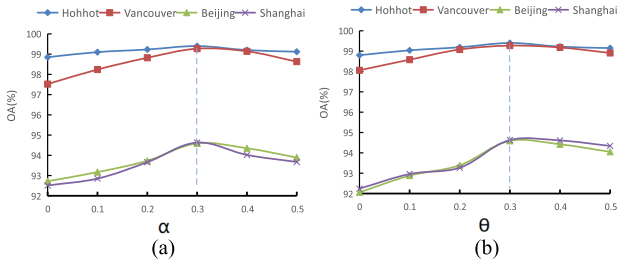



Fig. 7. (a) Results of different values of hyper-parameters $\alpha$. (b) Results of different values of hyper-parameters $\theta$.

Then, with $\alpha$ as the middle value 0.3, we vary $\theta$ from 0 to 0.5 with a stride of 0.1 and get the corresponding

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TABLE I
HYPER-PARAMETERS OF SSMU-NET

| Stage | Epoch | Operation | Input | Network | Output |
|---|---|---|---|---|---|
| Data Acquisition | —— | DWT | MS(W×H, 4) | —— | $Ms^{pre}$(0.5W×0.5H, 12)<br>$Mc^{pre}$(0.5W×0.5H, 4) |
| | | | PAN(4W×4H, 1) | —— | $Ps^{pre}$(2W×2H, 3)<br>$Pc^{pre}$(2W×2H, 1) |
| | 5 | Gm | MS(W×H, 4) | ResB[3×3, c]×4 | $Ms^{gan}$(0.5W×0.5H, 12) |
| | | Gp | PAN(4W×4H, 1) | | $Ps^{gan}$(2W×2H, 3) |
| | | Dm / Dp | $Ms^{pre}$(0.5W×0.5H, 12)<br>$Ms^{gan}$(0.5W×0.5H, 12)<br>$Ps^{pre}$(2W×2H, 3)<br>$Ps^{gan}$(2W×2H, 3) | Conv[3×3, c]<br>ResB[3×3, c]×3<br>Pooling<br>Sigmoid | [0,1] |
| Feature Traction | 30 | Conv | $Ms^{gan}$(0.5W×0.5H, 12)<br>MS(1W×1H, 4)<br>$Ps^{gan}$(2W×2H, 3)<br>PAN(4W×4H, 1) | Conv[3×3, c]<br>ResB[3×3, c]×3<br>Sigmoid(only $Ms^{gan}$, $Ps^{gan}$)<br>ResB[3×3, c]×2 | Ms(0.5W×0.5H, 8)<br>Ps(0.5W×0.5H, 8) |
| | | | $Mc^{pre}$(0.5W×0.5H, 4)<br>$Pc^{pre}$(2W×2H, 1) | Conv[3×3, c]<br>ResB[3×3, c]×5 | Mc(0.5W*0.5H, 8)<br>Pc(0.5W*0.5H, 8) |
| Mode Unification | 5 | EDm / EDp | Mc(0.5W×0.5H, 8)<br>Pc(0.5W×0.5H, 8)<br>Ms(0.5W×0.5H, 8)<br>Ps(0.5W×0.5H, 8) | ResB[3×3, c]×4 | $Mc^{u}$(0.5W×0.5H, 8)<br>$Pc^{u}$(0.5W×0.5H, 8)<br>$Ms^{u}$(0.5W×0.5H, 8)<br>$Ps^{u}$(0.5W×0.5H, 8) |
| Fusion Classification | 15 | Max | $Mc^{u}$(0.5W×0.5H, 8)<br>$Pc^{u}$(0.5W×0.5H, 8) | —— | $MPc^{u}$(0.5W×0.5H, 8) |
| | | Conv | $Ms^{u}$(0.5W×0.5H, 8)<br>$Ps^{u}$(0.5W×0.5H, 8)<br>$MPc^{u}$(0.5W×0.5H, 8) | concatenate<br>ResB[3×3, c]×5<br>Pooling<br>Fully Connected | Class |

results in Fig. 7(b). It can be seen that in each dataset, the performance is best when $\theta$ equals 0.3, and when $\theta$ is away from 0.3, OA decreases. If $\theta$ is very small, the learning tendency of the network is likely to lose the advantageous features of modes. If the value of $\theta$ is large, the result is not very terrible.

### D. Module Comparison Experiments

In this section, we take the Hohhot dataset as an example to verify the effectiveness of our proposed module, including producing style components by GAN, the STT module, and the MU module. For the fairness of our experiment, we do not change the parameters of any other parts except for the modules to be compared.

*1) Analysis of Producing Style Components by GAN:* We demonstrate the necessity of this operation by visualizing feature maps and comparing classification accuracy. As can be seen in Fig. 8, $Ms^{pre}$ and $Ps^{pre}$ contain much less information than $Mc^{pre}$ and $Pc^{pre}$, so their volume of information is not suitable for the traction in SST module. $Ms^{gan}$ and $Ps^{gan}$ have a greater volume of information, and focus on edge information. Therefore, they have both high-frequency semantic information and low-frequency detail information. In the subsequent ablation study, we also provided a quantitative comparison of classification accuracy between using this operation and directly using the results of the wavelet decomposition.

*2) Validation of SST Module:* We demonstrate the validation of the SST module by comparing the results without the process of the SST module. Similarly, we still adopt the method of controlling variables for our experiments.

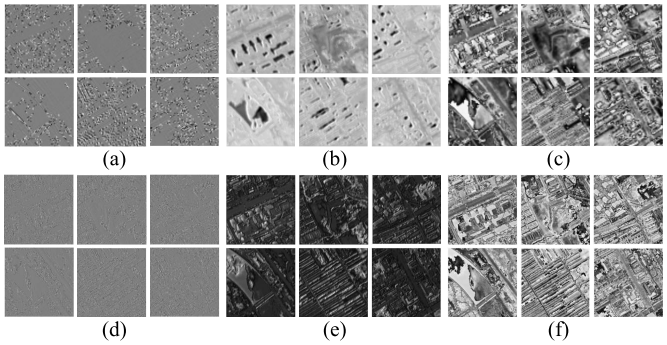



Fig. 8. We have selected six sets of images, each comprising six distinct forms. Images of each form are placed together. (a) $Ms^{pre}$, (b) $Ms^{gan}$, (c) $Mc^{pre}$, (d) $Ps^{pre}$, (e) $Ps^{gan}$, and (f) $Pc^{pre}$.

As can be seen in Table II, the results processed by the SST module are much higher than those obtained using only convolution. General convolution does not pay close attention to the style and content properties, causing confusion and redundancy in different feature spaces. However, the main function of the SST module is to achieve traction between different branches, so that the advantageous features from different modes can be given more attention. With the help of triplet loss, the SST module can obtain richer and purer feature representations.

*3) Validation of MU Module:* The role of the MU module is to map features representing different styles that are

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TABLE II
VALIDATION OF SST MODULE

| Process | Without SST | With SST |
|---|---|---|
| c1 | 98.07 | 98.59 |
| c2 | 93.71 | 97.03 |
| c3 | 95.79 | 97.54 |
| c4 | 90.56 | 97.98 |
| c5 | 96.52 | 99.50 |
| c6 | 96.35 | 98.73 |
| c7 | 96.54 | 97.11 |
| c8 | 93.29 | 96.93 |
| c9 | 96.79 | 98.85 |
| c10 | 94.65 | 95.91 |
| c11 | 90.04 | 97.96 |
| OA | 95.21 | 98.10 |
| AA | 94.76 | 97.83 |
| KAPPA | 94.64 | 97.87 |

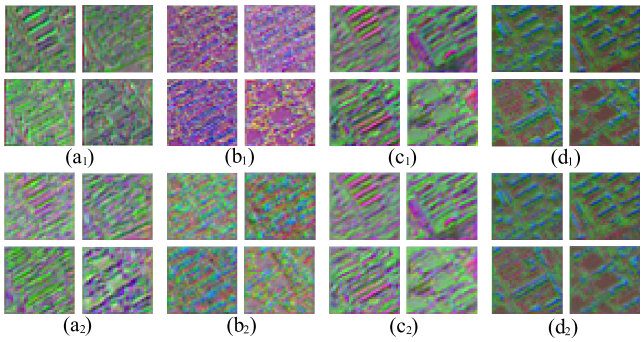



Fig. 9. We have selected four sets of images, each comprising eight distinct forms. Images of each form are placed together. (a1) *Ms*, (b1) *Ps*, (c1) *Mc*, and (d1) *Pc*. (a2) $Ms^u$, (b2) $Ps^u$, (c2) $Mc^u$, and (d2) $Pc^u$.

purified from different modes to a unified intermediate mode, in order to reduce the difficulty of fusion. Thus, we visually illustrate the process of feature changes by visualization. In the final ablation study, we also provide quantitative evidence to demonstrate the performance improvement brought by this module.

As can be seen in Fig. 9, the content of all features remains almost unchanged, but the presentation mode varies. Due to the fact that the SST module makes content features closer, there is little change after processing. Style features are close to each other with the changed concentration of colors, while the content structure presented by the color blocks is not destroyed.

*4) Ablation Study:* In this section, we take the Hohhot dataset as an example to analyze the contribution and effectiveness by combining different modules. We use Resnet-18 as the baseline. We abbreviate the three parts of the SS Strategy as Wavelet, GAN, and STT. The specific ablation study results are shown in Table III.

It can be seen in Table III that the OA of the model with style components generated by GAN improves by 0.95% from the baseline. However, without it, directly using STT has a counterproductive effect. This is because although the order of information varies, the results of wavelet decomposition still retain most of the information. However, the traction of STT

TABLE III
EXPERIMENTAL RESULTS OF EACH OPERATION ON HOHHOT DATASET

| Baseline | Wavelet | GAN | STT | MU | OA | AA | Kappa |
|---|---|---|---|---|---|---|---|
| ✓ | | | | | 93.70 | 93.45 | 92.87 |
| ✓ | ✓ | | | | 94.26 | 93.94 | 93.59 |
| ✓ | ✓ | ✓ | | | 95.21 | 94.76 | 94.64 |
| ✓ | ✓ | | ✓ | | 93.14 | 93.34 | 92.35 |
| ✓ | ✓ | ✓ | ✓ | | 98.10 | 97.83 | 97.87 |
| ✓ | ✓ | ✓ | ✓ | ✓ | **99.40** | **99.39** | **99.33** |

TABLE IV
EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON THE BEIJING DATASET

| Methods | SCPF-ResNet | SCCA-Net | AHF-Net | MPCA-Net | TRLF-Net | SSMU-Net |
|---|---|---|---|---|---|---|
| c1 | 84.07 | 85.94 | 88.10 | 91.57 | 90.83 | **93.74** |
| c2 | 80.79 | 55.80 | 80.17 | **92.71** | 84.27 | 89.44 |
| c3 | 87.20 | 77.39 | 82.17 | 90.43 | 89.81 | **93.63** |
| c4 | 78.05 | **95.91** | 87.43 | 89.76 | 82.82 | 94.11 |
| c5 | 59.13 | 61.40 | 66.02 | 86.29 | 70.09 | **87.93** |
| c6 | 90.21 | 90.81 | 91.13 | 90.57 | 91.65 | **94.60** |
| c7 | 84.98 | 91.90 | 85.29 | 90.46 | 90.94 | **94.83** |
| c8 | 89.40 | 77.14 | 85.23 | 92.48 | 91.11 | **94.96** |
| c9 | 69.34 | 54.53 | 80.63 | **89.96** | 84.52 | 87.74 |
| c10 | 53.14 | 75.48 | 79.67 | 87.27 | 82.09 | **93.55** |
| OA(%) | 86.24 | 84.29 | 85.95 | 90.26 | 89.20 | **94.59** |
| AA(%) | 77.63 | 76.63 | 76.54 | 90.15 | 80.44 | **92.45** |
| Kappa(%) | 79.08 | 76.86 | 78.65 | 89.72 | 83.65 | **91.82** |
| Param($n$) | $4.27\times10^6$ | $7.82\times10^6$ | $3.94\times10^6$ | $3.98\times10^6$ | $1.32\times10^7$ | $5.82\times10^6$ |
| Time($s$) | 4328.16 | 4486.23 | 3002.40 | 2049.36 | 2897.49 | 2187.44 |

on different orders of information destroys the information. The combination of GAN and STT has the highest contribution to the overall network, and its OA is increased by 3.84% over the baseline. The SS module achieves feature separation and purification, highlighting advantageous features. With the MU module, the OA of the model continues to achieve a 1.30% improvement. The MU module eliminates mode differences, making contributions to feature fusion.

### *E. Experimental Result on Beijing Dataset*

In this section, we use the Beijing dataset as an example to compare various methods and verify in detail the effectiveness and contribution of the proposed method. The corresponding quantitative and qualitative results of the Beijing dataset are shown in Table IV and Fig. 10.

SCCA-Net designed a global channel attention (GCA) module for MS and a local spatial attention (LSA) module for PAN to achieve feature enhancement. However, the accuracy of this method is very unstable for different categories. For c7 (which occupies half of the total quantity of labels), the accuracy is relatively high. For c2 (road) and c9 (red building) (only about 2% of the labels), the accuracy is not ideal. Therefore, the algorithm is more affected by sample imbalance.

Compared with SCCA-Net, SCPF-Net has significantly improved its accuracy in the c2 category (road). SCPF-Net designs an adaptive spatial attention (ASA) module and an adaptive channel attention (ACA) module to enhance the

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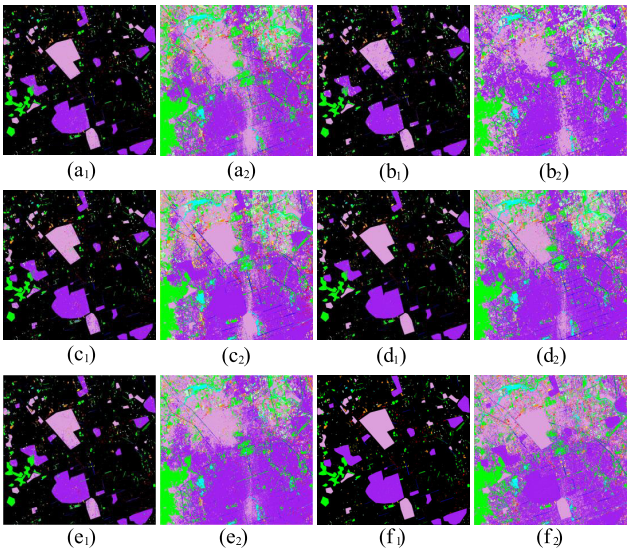



Fig. 10. Classification maps of different methods on the Beijing dataset. (Left) Image with ground truth and (right) overall image. (a1) and (a2) SCPF-ResNet. (b1) and (b2) SCCA-Net. (c1) and (c2) AHF-Net. (d1) and (d2) MPCA-Net. (e1) and (e2) TRLF-Net. (f1) and (f2) SSMU-Net.

spatial features of targets with different sizes. However, the interactive data fusion strategy (IDFS) reduces mode differences without strengthening the advantageous features in the data fusion stage. Therefore, for categories that are scarce and easily confused, such as c5 (blue building) and c10 (white building), the performance is not ideal. As a whole, the classification accuracy of this method needs to be improved.

AHF-Net proposes a new fusion strategy named AWIHS, which effectively reduces spectral distortion in multimodal data fusion. It also uses a feature pyramid network to recognize targets at multiple scales, improving the classification accuracy of most categories (such as c1, c4, c6, and c9). However, this approach does not focus on considering the utilization of the advantageous features of different modes.

Compared to the above methods, MPCA-Net improves OA by approximately 5%. In most categories, this method greatly improves classification accuracy, and the recognition performance for each category is stable. MPCA-Net proposes the ADR-SS to divide category areas surrounding the environment, so as to find the most suitable convolution kernel for a receptive field to carry on feature extraction. This method increases the stability of recognition and is very helpful in classifying categories in the complex surrounding environment.

Besides, TRLF-Net also has good classification performance. TRLF-Net designs a dual-branch attention sparse transfer module (DAST) to migrate advantage attributes of PAN and MS and proposed a module named RFGAL to solve the problem of large intraclass differences and small interclass differences.

Compared to previous methods, our SSMU-Net has the best overall performance. Our method achieves the highest accuracy except for c2 (road), c4 (lowvegetation), and c9 (red building). Especially, our accuracy improves greatly in c7 (dense high buildings), c8 (dense low buildings), and most nonbuilding categories. The information distribution of these categories in remote sensing images usually exhibits obvious style information. Our method makes good use of these features. We separate features from different attributes and extract them separately in different branches, highlighting style features. Furthermore, compared to the previous method, we first separate different style features, then eliminate mode differences, and finally extract features. This method to some extent avoids the loss of style features during the process of eliminating mode differences in the data fusion stage.

Finally, compared to various methods, the number of parameters in our network is slightly larger, because the SS strategy requires processing four branches separately. However, since our method is an end-to-end network and most modules can be seamlessly executed on GPUs, the overall runtime of our method is acceptable. In the future, we will optimize our multibranch strategy to make the overall method more efficient.

TABLE V
EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON THE HOHHOT DATASET

| Methods | SCPF-ResNet | SCCA-Net | AHF-Net | MPCA-Net | TRLF-Net | SSMU-Net |
|---|---|---|---|---|---|---|
| c1 | **99.69** | 98.84 | 99.22 | 99.61 | 99.12 | 99.56 |
| c2 | 97.21 | 94.72 | 94.35 | 96.95 | **98.10** | 97.85 |
| c3 | 97.30 | 95.37 | 96.25 | 96.13 | 98.83 | **99.06** |
| c4 | 98.48 | 96.24 | 96.81 | 97.63 | 99.26 | **99.68** |
| c5 | 97.53 | 98.09 | 98.52 | 97.84 | 99.76 | **99.83** |
| c6 | 97.64 | 96.12 | 97.23 | 96.20 | 99.40 | **99.88** |
| c7 | 98.54 | 98.84 | 98.10 | 98.88 | 99.19 | **99.40** |
| c8 | 98.65 | 96.57 | 97.83 | 97.47 | 99.39 | **99.94** |
| c9 | 97.96 | 97.63 | 96.22 | 96.78 | 98.98 | **99.15** |
| c10 | 97.61 | 96.93 | 96.15 | 96.79 | 98.96 | **99.74** |
| c11 | 97.63 | 98.11 | 95.69 | **99.35** | 98.55 | 99.24 |
| OA | 97.86 | 96.80 | 96.82 | 97.38 | 99.03 | **99.40** |
| AA | 97.78 | 96.95 | 96.64 | 97.60 | 98.93 | **99.39** |
| Kappa | 97.61 | 96.42 | 96.27 | 97.27 | 98.91 | **99.33** |
| Param($n$) | $4.27\times10^6$ | $7.82\times10^6$ | $3.94\times10^6$ | $3.98\times10^6$ | $1.32\times10^7$ | $5.82\times10^6$ |
| Time($s$) | 222.23 | 386.28 | 412.18 | 542.28 | 185.05 | 474.36 |

### *F. Experimental Result on Hohhot, Vancouver, and Shanghai Datasets*

We conducted the same experiment on three other datasets. The experimental results indicate that our SSMU-Net still has the highest OA, AA, and Kappa.

For the Hohhot dataset, the experimental results and images are shown in Table V and Fig. 11. Experimental results of various algorithms show good performance. However, our algorithm still exhibits a significant improvement. Particularly, we observe a significant increase in accuracy on c6 (sparse building), c9 (white building), and c10 (red building). These buildings have similar content features but have different style features. Our SS strategy can effectively capture the details of the style. For other categories, our algorithm's accuracy is also higher than most algorithms' or at the same level, reflecting good stability.

Given our slightly superior results on the Hohhot dataset compared to TRLF-Net, we introduced an additional experiment to compare the standard deviation. Table VI shows the standard deviations of TRLF-Net and SSMU-Net, which

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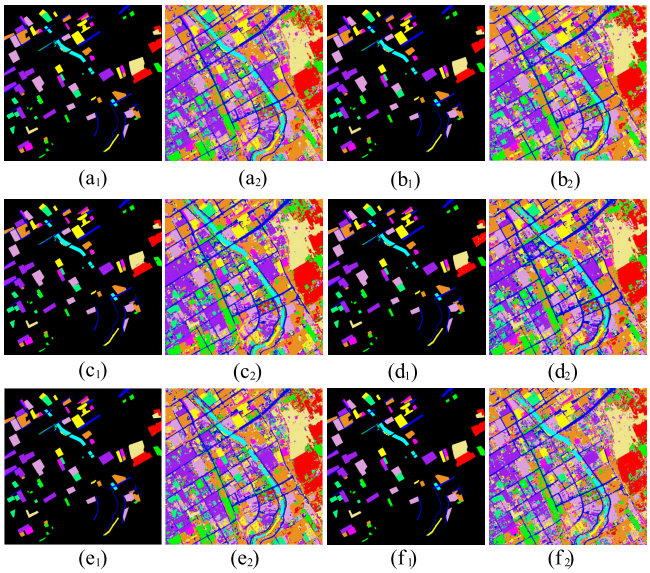



Fig. 11. Classification maps of different methods on the Hohhot dataset. (Left) Image with ground truth and (right) overall image. (a1) and (a2) SCPF-ResNet. (b1) and (b2) SCCA-Net. (c1) and (c2) AHF-Net. (d1) and (d2) MPCA-Net. (e1) and (e2) TRLF-Net. (f1) and (f2) SSMU-Net.

TABLE VI
STANDARD DEVIATIONS OF TRLF-NET AND SSMU-NET ON THE HOHHOT DATASET

| Methods | TRLF-Net | SSMU-Net |
|---|---|---|
| c1 | 0.26 | 0.15 |
| c2 | 0.27 | 0.20 |
| c3 | 0.18 | 0.13 |
| c4 | 0.22 | 0.14 |
| c5 | 0.11 | 0.06 |
| c6 | 0.16 | 0.04 |
| c7 | 0.21 | 0.15 |
| c8 | 0.23 | 0.15 |
| c9 | 0.19 | 0.21 |
| c10 | 0.30 | 0.09 |
| c11 | 0.23 | 0.21 |
| OA(%) | 0.15 | 0.09 |
| AA(%) | 0.11 | 0.08 |
| Kappa(%) | 0.15 | 0.09 |

are the results of ten experiments. Our algorithm exhibits lower standard deviations than TRLF-Net across nearly all categories, verifying the stability of our algorithm. In fact, in other datasets, our algorithm also reflects good stability.

Table VII and Fig. 12 present the performance evaluation of each algorithm using the Vancouver dataset. Certain algorithms exhibit suboptimal performance when dealing with c6 (road) and c8 (boat). They are often intertwined with buildings and have patches similar to adjacent samples of different categories, making it hard for conventional attention mechanisms to classify. MPCA-Net's center pixel migration (CPM) strategy is more targeted at this issue, performing well in these two categories. Our algorithm considers the distinctive style of each category, thereby achieving the highest level of stability among all algorithms. Although our performance on c1 and c8 shows slight deviations from that of MPCA-Net, our algorithm outperforms its counterparts in most categories, particularly on buildings.

TABLE VII
EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON THE VANCOUVER DATASET

| Methods | SCPF-ResNet | SCCA-Net | AHF-Net | MPCA-Net | TRLF-Net | SSMU-Net |
|---|---|---|---|---|---|---|
| c1 | 91.96 | 89.19 | 89.30 | **98.88** | 97.26 | 98.83 |
| c2 | 90.02 | 87.50 | 90.53 | 97.67 | 96.97 | **99.60** |
| c3 | 94.59 | 92.61 | 93.84 | 98.81 | 98.89 | **99.49** |
| c4 | 88.58 | 83.69 | 88.14 | 98.21 | 95.68 | **98.25** |
| c5 | 99.44 | 99.65 | 99.76 | 97.34 | 99.81 | **99.90** |
| c6 | 66.10 | 54.45 | 72.46 | 97.29 | 94.22 | **98.47** |
| c7 | 96.06 | 86.48 | 91.0 | 98.10 | 98.39 | **98.90** |
| c8 | 81.34 | 46.05 | 72.68 | **97.77** | 96.90 | 97.02 |
| OA(%) | 92.20 | 89.53 | 92.65 | 97.45 | 98.13 | **99.28** |
| AA(%) | 88.51 | 79.95 | 87.22 | 98.01 | 97.27 | **98.81** |
| Kappa(%) | 89.71 | 86.22 | 90.33 | 97.16 | 97.55 | **99.05** |
| Param($n$) | $4.27\times10^6$ | $7.82\times10^6$ | $3.94\times10^6$ | $3.98\times10^6$ | $1.32\times10^7$ | $5.82\times10^6$ |
| Time($s$) | 101.95 | 111.25 | 100.86 | 107.39 | 175.82 | 129.34 |

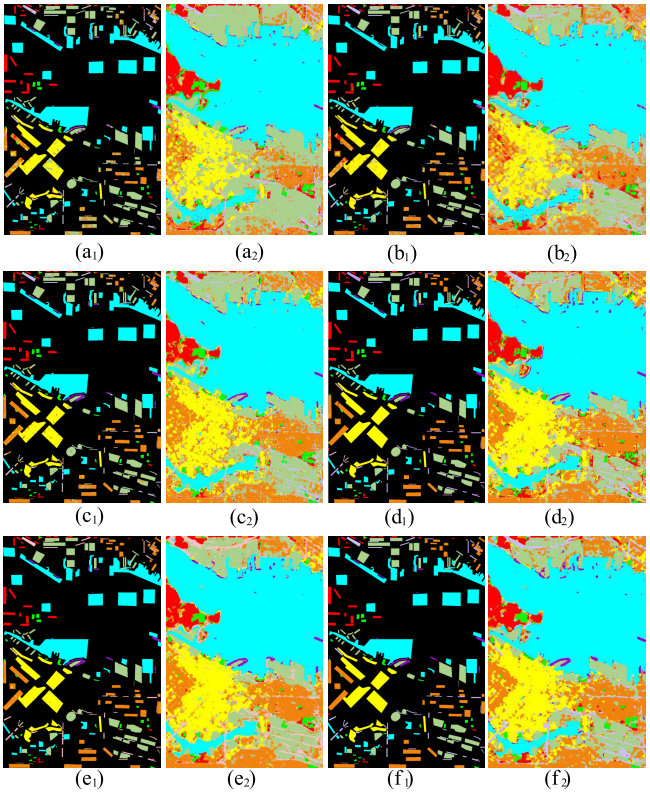



Fig. 12. Classification maps of different methods on the Vancouver dataset. (Left) Image with ground truth and (right) overall image. (a1) and (a2) SCPF-ResNet. (b1) and (b2) SCCA-Net. (c1) and (c2) AHF-Net. (d1) and (d2) MPCA-Net. (e1) and (e2) TRLF-Net. (f1) and (f2) SSMU-Net.

For the Shanghai dataset, our algorithm achieves significant improvements in all three indicators. The specific results are shown in Table VIII and Fig. 13. Compared to other algorithms, our algorithm achieves an accuracy of over 90% for each class for the first time. Our algorithm has significantly higher accuracy than other algorithms in the three categories of c7 (dense high building), c8 (dense low building), and c10 (white building). This is because our algorithm considers more style features, making it easier to distinguish similar buildings that are easily confused. In addition, it can be

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TABLE VIII
EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON THE SHANGHAI DATASET

| Methods | SCPF-ResNet | SCCA-Net | AHF-Net | MPCA-Net | TRLF-Net | SSMU-Net |
|---|---|---|---|---|---|---|
| c1 | 86.37 | 83.49 | 83.71 | 92.37 | 89.26 | **92.71** |
| c2 | 76.55 | 78.79 | 52.90 | 93.68 | 91.99 | **97.09** |
| c3 | **95.68** | 89.55 | 77.90 | 91.28 | 88.83 | 90.81 |
| c4 | **94.38** | 93.62 | 64.76 | 88.91 | 81.90 | 93.54 |
| c5 | 88.09 | 71.74 | 80.28 | 89.27 | 88.93 | **93.48** |
| c6 | 86.24 | 77.96 | 76.67 | 87.42 | 86.07 | **92.46** |
| c7 | 91.11 | 85.44 | 76.66 | 90.10 | 92.50 | **95.94** |
| c8 | 85.14 | 84.88 | 83.01 | 85.49 | 87.91 | **92.45** |
| c9 | 77.58 | 92.35 | 65.61 | **94.57** | 93.21 | 91.19 |
| c10 | 94.74 | 84.26 | 93.61 | 95.87 | 96.78 | **97.88** |
| OA(%) | 89.03 | 84.40 | 78.38 | 89.83 | 90.64 | **94.63** |
| AA(%) | 87.59 | 84.51 | 75.51 | 90.70 | 88.95 | **93.76** |
| Kappa(%) | 82.09 | 73.69 | 65.56 | 89.69 | 84.60 | **91.77** |
| Param($n$) | $4.27\times10^6$ | $7.82\times10^6$ | $3.94\times10^6$ | $3.98\times10^6$ | $1.32\times10^7$ | $5.82\times10^6$ |
| Time($s$) | 2967.96 | 4937.96 | 3100.89 | 2317.10 | 1187.00 | 2230.86 |

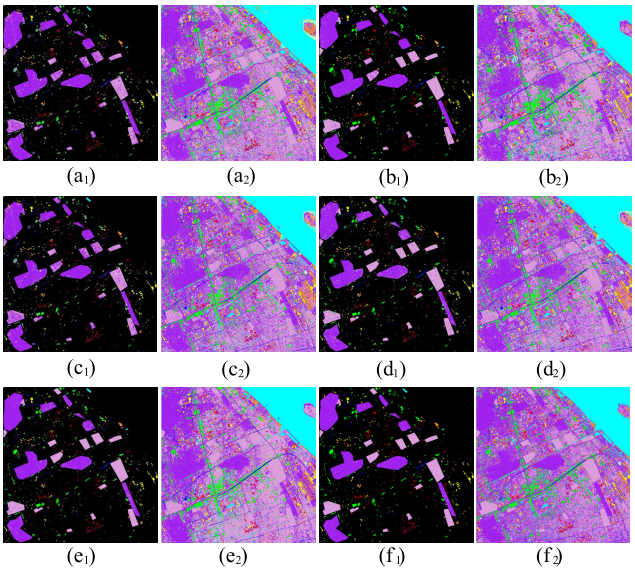



Fig. 13. Classification maps of different methods on the Shanghai dataset. (Left) Image with ground truth and (right) overall image. (a1) and (a2) SCPF-ResNet. (b1) and (b2) SCCA-Net. (c1) and (c2) AHF-Net. (d1) and (d2) MPCA-Net. (e1) and (e2) TRLF-Net. (f1) and (f2) SSMU-Net.

observed that our model exhibits high balance, as the accuracy is consistently high across all classes. This is because our model does not directly rely on pansharpening algorithms to eliminate mode differences during data fusion. Instead, it utilizes encoder–decoder models and perceptual loss after feature fusion to gradually map features extracted from different modes to similar modes. This approach avoids spectral distortion and achieves a more gentle processing effect.

## V. CONCLUSION

In this article, we proposed an SSMU-Net for the classification task of multimodal remote sensing images. Experiments on four datasets verified our proposed method's effectiveness of fully utilizing multimodal advantageous features contributing to classification. SSMU-Net constructs ordered novel network modules from the perspective of unique styles of different modes. This method can separate information from paired remote sensing images and optimize them toward the task while eliminating the problem of mode differences. The SS strategy achieves the separation and traction of content and style features from remote sensing data of different modes, makes the extraction of different features more pure, and highlights the unique advantages of modes. The MU module reduces mode differences while preserving semantic information, realizing better fusion of features from different data sources. However, this style-based classification method requires processing multiple branches separately, resulting in a slightly larger overall parameter count. In the near future, we will optimize our method to improve efficiency while maintaining good performance.

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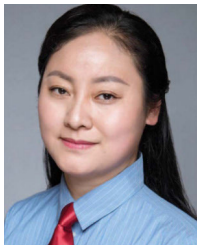
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